BUILDCHAIN: BLOCKCHAIN-BASED CONSTRUCTION MATERIAL AUTHENTICATION AND TRACEABILITY SYSTEM

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

Use Case Report

submitted by

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2026-27

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CERTIFICATE

This is to certify that the Use Case report entitled **"BuildChain: Blockchain-Based Construction Material Authentication and Traceability System"** that is being submitted by **Katakam Sirivalli(23501A0566)** , as part of Assignment-2 for the **Blockchain Technology(23CS4701B)** course in **4-1** during the academic year **2026-27**.

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1. INTRODUCTION

1.1. Overview of Blockchain Technology

Blockchain technology is a decentralized and distributed ledger system that records transactions across multiple nodes in a network, providing transparency, security, and tamper-resistant data storage. Unlike traditional centralized databases, blockchain enables multiple participants to maintain and verify a shared transaction history without depending entirely on a single central authority. Each block contains transaction data and a cryptographic hash that links it to the previous block, making unauthorized modification of recorded information detectable. Originally introduced for cryptocurrency transactions, blockchain technology has expanded into several domains, including finance, healthcare, supply chain management, digital identity, and product authentication. Its features, such as decentralization, cryptographic verification, consensus mechanisms, and smart contracts, make it suitable for applications that require reliable records and accountability among multiple stakeholders.

1.2. Relevance of Blockchain in Construction Material Authentication and Traceability Systems

In the construction industry, materials such as cement, steel, bricks, and other building products pass through several stages, including manufacturing, quality certification, inspection, distribution, transportation, and delivery to construction sites. Traditional material-tracking systems often rely on paper-based certificates or separate databases maintained by manufacturers, suppliers, distributors, and contractors. These practices can create difficulties in verifying material authenticity, checking certification records, tracking material movement, and identifying unauthorized changes to recorded information. Such challenges may increase the risk of counterfeit materials entering the supply chain and reduce transparency among stakeholders.

BuildChain: Blockchain-Based Construction Material Authentication and Traceability System proposes the use of blockchain technology to maintain a shared, tamper-resistant record of construction materials throughout their supply chain. Each material batch can be assigned a unique identification number and associated QR code. Important information, including manufacturer details, batch numbers, certificate references, inspection results, ownership transfers, and delivery records, can be recorded or referenced through blockchain transactions. Authorized stakeholders can scan the QR code to retrieve the material's registered history and verify its recorded status.

By leveraging smart contracts, BuildChain can control material registration, authorized certificate updates, inspection recording, and material transfers according to predefined rules. For example, a manufacturer can register a steel batch, an authorized inspector can record its inspection result, and subsequent suppliers or distributors can record transfers until the material reaches the construction site. Each authorized transaction

contributes to an auditable history, helping stakeholders identify inconsistencies and verify the sequence of recorded events.

Furthermore, blockchain can improve coordination between manufacturers, suppliers, distributors, contractors, and quality inspectors by providing access to a common record rather than requiring each participant to rely exclusively on separate internal records. Cryptographic hashes can also be used to check whether linked quality certificates and inspection documents have been altered after their registration. This supports document integrity and improves the reliability of material verification.

The proposed BuildChain system aims to strengthen transparency, accountability, and traceability in construction material management. However, blockchain alone cannot guarantee that a physical material is genuine or that the original information entered into the system is correct. Reliable authentication therefore depends on trusted manufacturers, authorized inspectors, valid certification processes, and suitable physical identification mechanisms. By combining these verification procedures with blockchain's tamper-resistant recordkeeping and smart-contract capabilities, BuildChain offers a practical approach to improving construction material authentication and supply-chain traceability.

2. BACKGROUND

The construction industry relies on various materials, including cement, steel, bricks, and other building products, to ensure the safety, durability, and quality of infrastructure. These materials pass through multiple stages, from manufacturing and quality certification to distribution, transportation, and delivery at construction sites. Traditional construction material management often depends on paper-based certificates, manual verification, and centralized databases maintained by different stakeholders. These practices can lead to difficulties in verifying material authenticity, tracking material movement, and maintaining reliable quality records. The following are some key challenges associated with existing construction material management systems.

2.1. Counterfeit Construction Materials

Counterfeit construction materials may enter the supply chain through unauthorized manufacturers, suppliers, or distributors. Such materials may not meet the required quality standards and can compromise structural safety and durability. Without a reliable verification mechanism, contractors may find it difficult to distinguish genuine materials from counterfeit products.

2.2. Forged Quality Certificates

Construction materials are often accompanied by quality certificates, test reports, and compliance documents. These documents may be forged, duplicated, or associated with the wrong material batch. Verifying the authenticity of certificates is essential to ensure that materials meet the required construction standards.

2.3. Lack of Supply-Chain Traceability

Materials pass through several stakeholders before reaching construction sites. When manufacturers, suppliers, distributors, and contractors maintain separate records, tracking a material batch from its origin to its final destination becomes difficult. This can delay investigations when material-related issues occur.

2.4. Lack of Transparency Among Stakeholders

Different participants may maintain separate databases and records, resulting in inconsistent information about material origin, certification, inspection, and ownership. The absence of a shared record can create disputes and reduce trust among supply-chain participants.

2.5. Unauthorized Modification of Records

Centralized databases and paper-based records may be vulnerable to unauthorized

changes, loss, or manipulation. Altered batch details, inspection results, or certificate information can make it difficult to establish a reliable history of construction materials.

2.6. Inefficient Material Verification

Contractors and site engineers may need to contact manufacturers or suppliers to verify material details and quality certificates. Manual verification can consume time, delay construction activities, and complicate material acceptance procedures.

2.7. Difficulty in Tracking Material Transfers

Construction materials may change ownership several times during distribution. If transfer records are incomplete or inconsistent, it becomes difficult to identify the current responsible party and verify the sequence of material movement.

2.8. Poor Inspection and Quality Record Management

Inspection results and laboratory test reports are important for evaluating construction material quality. When these records are maintained separately, retrieving and comparing inspection information can be challenging. This may also complicate quality audits and investigations.

2.9. Risk of QR Code Duplication and Product Substitution

QR codes can simplify material identification, but a copied code may be attached to a counterfeit or substituted product. Therefore, QR-based verification alone is insufficient. The system must also verify the registered material identity, authorized transactions, and associated history.

2.10. Limited Auditability and Accountability

When a material-related problem is discovered at a construction site, stakeholders may need to determine its origin, inspection history, and movement through the supply chain. Incomplete records can make investigations difficult and delay the identification of responsible parties.

Due to these challenges, there is a need for a reliable and transparent construction material authentication and traceability system. **BuildChain: Blockchain-Based Construction Material Authentication and Traceability System** proposes using blockchain technology to maintain a shared, tamper-resistant history of material registration, certification, inspection, and transfer. Smart contracts can control authorized transactions, while QR codes enable convenient access to registered material information. By improving traceability, record integrity, and accountability among stakeholders, BuildChain aims to reduce opportunities for record manipulation and support more reliable construction material verification.

3. BLOCKCHAIN BASICS

Blockchain technology is a distributed ledger system that records transactions across multiple nodes, providing transparency, security, and tamper-resistant recordkeeping. Unlike traditional centralized databases, blockchain enables multiple authorized participants to maintain and verify a shared transaction history. Its decentralized architecture, cryptographic security, and smart contract capabilities make it useful in applications involving multiple stakeholders who need to share reliable information. In **BuildChain: Blockchain-Based Construction Material Authentication and Traceability System**, these features support the registration, certification, inspection, and tracking of construction materials throughout the supply chain.

The following sections explain the fundamental blockchain concepts relevant to the proposed system:

3.1. Decentralization

Decentralization refers to distributing the maintenance and verification of records across multiple network participants rather than relying entirely on a single central authority. In a traditional construction material management system, a single organization may control the database containing material details, certificates, and transfer records.

In BuildChain, authorized participants such as manufacturers, suppliers, quality inspectors, distributors, and contractors can participate in a shared blockchain network. This allows them to verify material-related transactions using a common record.

For example, when a manufacturer registers a steel batch, authorized suppliers and contractors can subsequently verify its registered information and transfer history. This reduces exclusive dependence on one organization's internal records and improves coordination among supply-chain participants.

3.2. Immutability

Immutability is the property that makes recorded blockchain transactions difficult to alter without detection. Blockchain uses cryptographic hashes to link blocks, while its consensus mechanism helps participating nodes agree on valid transactions.

In BuildChain, events such as material registration, certificate references, inspection results, and ownership transfers can be recorded on the blockchain. Once confirmed, these records provide a tamper-resistant history that helps authorized participants identify inconsistencies or unauthorized changes.

For example, if a steel batch is registered and subsequently transferred through several suppliers, the blockchain maintains the recorded sequence of transactions. This supports auditing and makes it more difficult to conceal changes to the material's recorded history.

3.3. Transparency

Transparency allows authorized participants to inspect and verify relevant blockchain transactions. Depending on the network design and access permissions, participants can view the material information and transaction history they are authorized to access.

In BuildChain, manufacturers, inspectors, suppliers, and contractors can verify information such as material batch identifiers, certificate references, inspection status, and transfer records. A contractor can scan a material's QR code to access its registered details and verify the corresponding blockchain record.

This improves visibility throughout the supply chain and helps stakeholders identify missing records or inconsistencies. Since construction-related information may be commercially sensitive, a permissioned blockchain can restrict access to authorized participants rather than exposing all records publicly.

3.4. Smart Contracts

Smart contracts are programs deployed on a blockchain that execute predefined operations when specified conditions are satisfied. They help automate transactions, enforce access permissions, and reduce reliance on manual processing.

In BuildChain, smart contracts can manage construction material registration, certificate references, inspection records, and transfers between authorized stakeholders.

For example, a smart contract can verify whether the sender is authorized to transfer a registered material batch and update its recorded ownership when the transaction satisfies the required conditions. It can also prevent unauthorized users from modifying protected material records.

The main smart contract functions may include:

- registerMaterial() – registers a new material batch.
- registerCertificate() – links a certificate reference or hash to the batch.
- recordInspection() – records an authorized inspection result.
- transferMaterial() – records an authorized transfer.
- verifyMaterial() – retrieves and checks the registered material status.
- getMaterialHistory() – provides access to the recorded transaction history.

These functions improve consistency and traceability by applying predefined rules to material-related transactions.

3.5. Consensus Mechanisms

Consensus mechanisms are protocols through which blockchain network participants agree on the validity and ordering of transactions. They help maintain a consistent ledger across participating nodes and protect the network against invalid transaction submissions.

Common consensus mechanisms include Proof of Work (PoW), which uses computational work to validate blocks; Proof of Stake (PoS), which selects validators according to a stake-based mechanism; and other protocols designed for particular network

environments.

For BuildChain, a permissioned blockchain is suitable because participating organizations are identified and authorized. A permissioned network can use a consensus protocol appropriate for its membership, performance requirements, and governance model. This supports coordinated transaction validation without requiring every participant to use a public, permissionless network.

The selected consensus mechanism should ensure that material registration, inspection, and transfer transactions are consistently recorded by the authorized network participants.

3.6. Cryptographic Security

Cryptographic techniques protect blockchain transactions and help verify data integrity. Public-key cryptography allows users to sign transactions with their private keys, while other participants can verify the signatures using the corresponding public keys. Cryptographic hash functions generate fixed-length representations of data, allowing changes to documents to be detected by comparing their hashes.

In BuildChain, digital signatures can help verify that a transaction was submitted by the account associated with an authorized participant. Hashes can also be stored on the blockchain to verify the integrity of quality certificates, laboratory reports, and inspection documents maintained off-chain.

For example, when a quality certificate is registered, its cryptographic hash can be stored with the material batch record. If the certificate is later modified, calculating its hash again and comparing it with the recorded hash can reveal the change.

However, cryptographic verification does not independently prove that the original certificate is genuine or that the physical material meets the required quality standards. These aspects require trusted manufacturers, authorized inspectors, and reliable testing procedures.

By combining decentralization, immutability, transparency, smart contracts, consensus mechanisms, and cryptographic security, blockchain provides the technical foundation for BuildChain. These features support a shared and tamper-resistant record of construction material registration, certification, inspection, and movement. Consequently, the proposed system aims to improve material traceability, record integrity, and accountability among construction supply-chain participants.

4. USE CASE OVERVIEW

The use case for **BuildChain: Blockchain-Based Construction Material Authentication and Traceability System** aims to improve the authenticity verification, certification management, and traceability of construction materials by leveraging blockchain technology. The system provides a shared, tamper-resistant record of material registration, quality inspection, certification, and movement across the supply chain. By integrating smart contracts and QR-code-based identification, BuildChain enables authorized stakeholders to verify material details and track material batches from manufacturers to construction sites.

4.1. Objectives

The primary objective of BuildChain is to develop a blockchain-based platform for verifying construction material authenticity and maintaining reliable records throughout the supply chain.

A major challenge in construction material management is the possibility of counterfeit materials entering the market. BuildChain addresses this issue by recording material identification details and manufacturer information on the blockchain. This helps stakeholders verify whether the material's registered identity and history are consistent with the information provided.

Another objective is to improve the reliability of quality certification. Construction materials are associated with certificates, laboratory reports, and inspection records. By recording certificate references and cryptographic hashes on the blockchain, the system can help detect changes to registered documents and support verification of their integrity.

The system also aims to improve material traceability by recording transfers between manufacturers, suppliers, distributors, and contractors. Each authorized transaction contributes to a chronological history, making it easier to identify the recorded source and movement of a material batch.

Automation is another important objective. Smart contracts can enforce predefined rules for material registration, authorized transfers, inspection recording, and status updates. This reduces dependence on manual recordkeeping and helps prevent unauthorized transactions within the system.

Furthermore, BuildChain aims to improve transparency and accountability among supply-chain participants. Authorized users can access relevant material information through a web application and verify records using a unique material ID or QR code. This can simplify material verification at construction sites and support more efficient quality audits.

Overall, the system aims to strengthen record integrity, reduce opportunities for document manipulation, improve supply-chain visibility, and support reliable construction material verification.

4.2. Scope of the System

The proposed blockchain-based construction material authentication and traceability system focuses on registering, verifying, and tracking construction materials throughout their supply chain.
The system includes the following components:
1. **Manufacturers:** Register construction materials, assign unique batch identifiers, and provide certificate references and product details.
2. **Quality Inspectors:** Record inspection results and link relevant quality reports to registered material batches.
3. **Suppliers and Distributors:** Record material receipt and authorized transfers, maintaining the movement history of each batch.
4. **Contractors and Construction Sites:** Verify material information, inspect recorded certification and transfer history, and confirm receipt of materials.
5. **Web Application:** Provides an interface for authorized users to register materials, manage records, transfer materials, and verify authenticity.
6. **Blockchain Network:** Maintains the shared, tamper-resistant transaction history and executes smart contracts that enforce predefined rules.
7. **QR Code Verification:** Enables users to retrieve the registered material information associated with a unique identifier.
The system focuses on the verification of recorded material information and its history. Physical material testing and the initial confirmation of authenticity remain the responsibility of authorized manufacturers, laboratories, and inspectors.

4.3. System Architecture

The architecture of BuildChain consists of three major layers: the User Layer, Web Application Layer, and Blockchain Layer. These layers work together to support material registration, certification, inspection, transfer, verification and is illustrated in
Figure 4.1: Architecture of BuildChain—Blockchain-Based Construction Material Authentication and Traceability System

4.3.1. User Layer

The User Layer consists of the stakeholders who interact with the proposed system.
Manufacturers:
- Register construction materials and create unique batch records.
- Provide manufacturer details, batch numbers, and certificate references.
- View and manage registered material information.

Quality Inspectors:
- Verify material quality using appropriate inspection procedures.
- Record inspection results and certification references.
- Update inspection status through authorized transactions.

Suppliers and Distributors:

- Record the receipt of material batches.
- Transfer materials to authorized participants.
- View the registered source and movement history.

Contractors:

- Verify material authenticity before accepting materials at construction sites.
- Scan QR codes to retrieve material details.
- Review certificate references, inspection status, and transfer history.

4.3.2. Web Application Layer

The Web Application Layer acts as the interface between users and the blockchain network. It enables stakeholders to interact with the system without directly executing blockchain operations manually.

User Interface:

- Provides material registration and verification forms.
- Enables users to view material details and transaction history.
- Supports QR-code generation and scanning.
- Displays certificate references, inspection status, and transfer records.

Material Management System:

- Handles user registration and authentication.
- Checks user permissions before allowing operations.
- Facilitates communication between the frontend and smart contracts.
- Displays blockchain transaction results and material status.

QR Code Module:

- Generates a unique QR code linked to the material's registered identifier.
- Allows authorized users to retrieve the corresponding blockchain record.
- Helps identify inconsistencies between the presented material information and the registered record.

4.3.3. Blockchain Layer

The Blockchain Layer provides the shared transaction record, cryptographic verification, and smart contract execution required by BuildChain.

Smart Contract Layer:

The smart contracts manage the registration and tracking of construction material batches.

- **Material Registry Contract:** Stores material identifiers, manufacturer references, batch information, and registration status.
- **Certificate Management:** Records certificate references or hashes for integrity verification.
- **Inspection Management:** Records inspection outcomes submitted by authorized inspectors.
- **Material Transfer Management:** Records authorized transfers between supply-chain participants.

- **Verification Functions:** Retrieve material details and the chronological history of recorded events.

Blockchain Network:

- **Distributed Ledger:** Maintains a shared history of confirmed material-related transactions among participating nodes.
- **Tamper-Resistant Records:** Helps detect unauthorized changes to previously recorded transactions.
- **Transaction Validation:** Uses the network's consensus mechanism to validate and order transactions.
- **Access Control:** Restricts sensitive operations to authorized participant accounts according to predefined rules.

The blockchain stores essential material identifiers, transaction events, statuses, timestamps, and certificate hashes or references. Large documents, such as complete laboratory reports and inspection certificates, can be stored off-chain, with their cryptographic hashes recorded on the blockchain.

By integrating these three layers, BuildChain provides a structured approach to construction material authentication and traceability. The User Layer enables stakeholder interaction, the Web Application Layer manages requests, and the Blockchain Layer maintains the shared transaction history and enforces predefined rules through smart contracts. This architecture supports verification of material records from registration to delivery at construction sites. his layer ensures security, transparency, and immutability of warranty transactions.

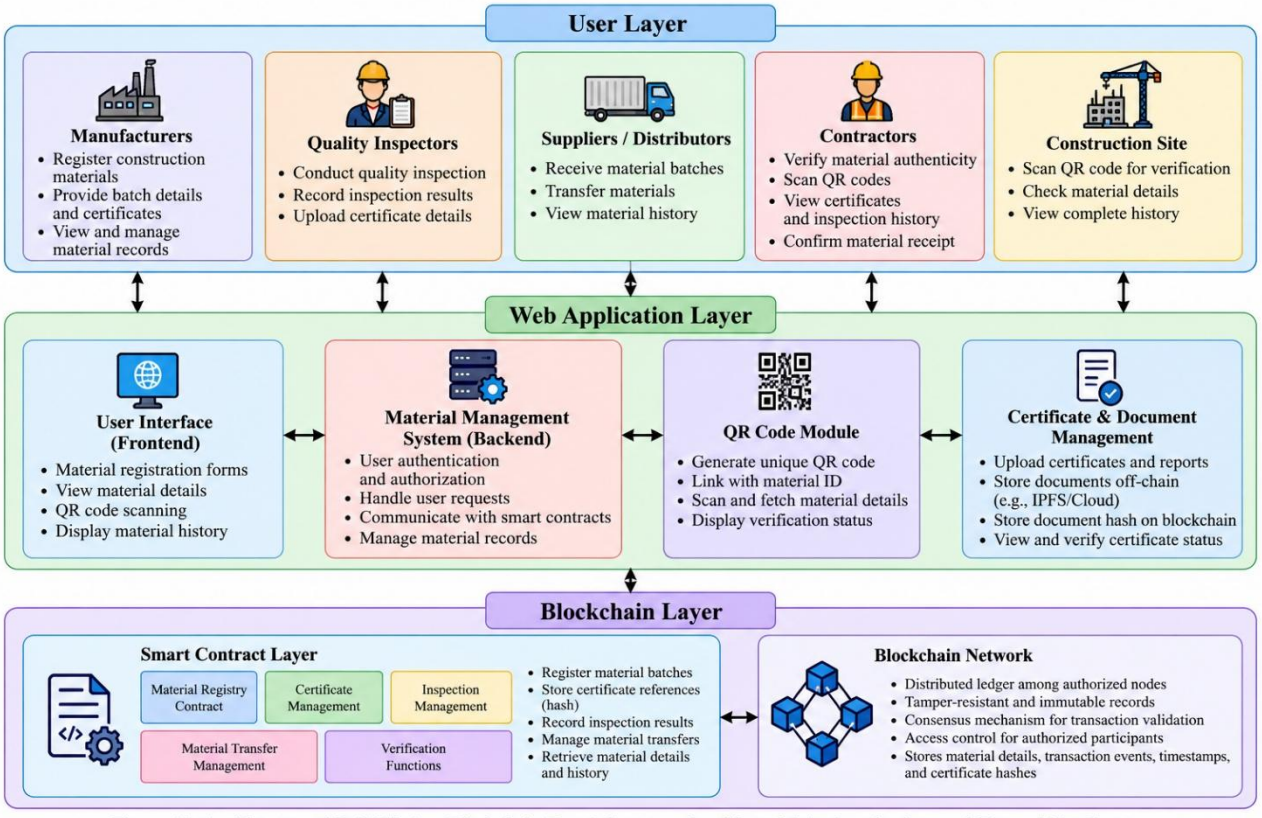


Figure 4.1: Architecture of BuildChain—Blockchain-Based Construction Material Authentication and Traceability System

5.IMPLEMENTATION

5.1. Setting Up the Blockchain Environment

To implement BuildChain: Blockchain-Based Construction Material Authentication and Traceability System, a suitable blockchain development environment is required to develop, deploy, and test smart contracts for construction material registration and traceability.

The proposed system uses an Ethereum-compatible blockchain platform and Solidity for smart contract development. Remix IDE can be used to write, compile, and deploy the contracts, while Ganache provides a local blockchain environment for testing transactions. MetaMask is used to manage blockchain accounts and authorize transactions. The frontend can be developed using HTML, CSS, and JavaScript, with Ethers.js facilitating communication between the web application and the blockchain.

Each construction material batch is identified by a unique Material ID. The system is designed to record material registration details, certificate references, inspection results, authorized transfers, and delivery events. QR codes provide a convenient way to retrieve the corresponding material information.

The proposed technology stack consists of:

- **Blockchain Platform:** Ethereum-compatible blockchain
- **Smart Contract Language:** Solidity
- **Development Environment:** Remix IDE
- **Local Blockchain:** Ganache
- **Wallet:** MetaMask
- **Frontend:** HTML, CSS, and JavaScript
- **Blockchain Integration:** Ethers.js
- **Material Identification:** QR-code generation and scanning

After configuring the environment, the smart contracts can be compiled, deployed, and tested using sample construction material records.

5.2. Defining Smart Contracts

5.2.1. Writing the Smart Contract for Warranty Management

The Material Registry Smart Contract manages the registration and verification of construction material batches. It assigns a unique identifier to each batch and maintains the material information defined in the contract.

A material record may contain the Material ID, manufacturer address, material type, batch number, registration timestamp, certificate reference or hash, inspection status, current custodian, and material status. The actual fields stored depend on the Solidity contract implementation.

Mappings can be used to associate Material IDs with their corresponding records and to identify authorized participants. Smart contract functions validate registration requests, inspection updates, and material transfers according to predefined access-control rules.

The blockchain maintains records submitted through confirmed transactions, providing a tamper-resistant history of material-related activities. This helps authorized stakeholders review the recorded history of construction materials throughout the supply chain.

```
struct Material {
    uint256 id;
    address manufacturer;
    string materialType;
    string batchNumber;
    uint256 registeredAt;
    bytes32 certificateHash;
    string inspectionStatus;
    address currentCustodian;
    string materialStatus;
}

mapping(address => bool) internal registeredManufacturers;
mapping(uint256 => Material) internal materials;
mapping(uint256 => bool) internal materialExists;
mapping(uint256 => address[]) internal materialHistory;
```

5.2.2 Registering Manufacturers Authorized Participants

To prevent unauthorized material registration and record updates, participants must be assigned appropriate permissions before performing restricted operations.Manufacturers are responsible for registering material batches and providing the associated material details. Authorized quality inspectors record inspection results, while suppliers, distributors, and contractors participate in material transfers and delivery confirmation according to their assigned roles.

The system validates the participant's permissions before executing restricted smart contract functions. It can also prevent duplicate Material IDs and reject transactions that do not satisfy the contract's predefined conditions.

For example, an authorized manufacturer can register a steel batch with the identifier STEEL-2026-001. Subsequent transactions can reference this identifier to associate

```
mapping(address => bool) internal registeredManufacturers;

function registerManufacturer() external {
    require(
        !registeredManufacturers[msg.sender],
        "Already registered"
    );

    registeredManufacturers[msg.sender] = true;
}
```

certificate information, inspection results, and transfer events with the same material batch.

5.2.3 Registering Construction Materials

After authorization, the manufacturer registers a construction material batch through the web application.The registration function checks the submitted information and verifies whether the Material ID already exists. If the request satisfies the required conditions, the contract creates the material record and generates a blockchain transaction.An example material record is shown below:

- Material ID: STEEL-2026-001
- Material Type: TMT Steel
- Batch Number: B1001
- Manufacturer ID: M001
- Certificate Reference: CERT-001
- Initial Status: Registered

The registration transaction establishes the recorded identity of the material batch and provides a reference for subsequent operations. Authorized users can use the Material ID to retrieve its registered information and associated transaction history.

```
uint256 private nextMaterialId = 1;

function registerMaterial(
    string memory materialType,
    string memory batchNumber,
    bytes32 certificateHash
) external returns (uint256) {
    require(
        registeredManufacturers[msg.sender],
        "Not a registered manufacturer"
    );

    uint256 materialId = nextMaterialId++;

    materials[materialId] = Material({
        id: materialId,
        manufacturer: msg.sender,
        materialType: materialType,
        batchNumber: batchNumber,
        registeredAt: block.timestamp,
        certificateHash: certificateHash,
        inspectionStatus: "Pending",
        currentCustodian: msg.sender,
        materialStatus: "Registered"
    });

    materialExists[materialId] = true;
```

```
    materialHistory[materialId].push(msg.sender);

    return materialId;
}
```

5.2.4 Registering Quality Certificates

Construction materials may be accompanied by quality certificates, laboratory reports, and inspection documents. BuildChain is designed to associate these documents with the corresponding material batches. If certificate-hash functionality is implemented, the system can record a cryptographic hash of the certificate on the blockchain. The complete document may be maintained separately using a suitable file-storage system. During verification, the system calculates the hash of the retrieved document and compares it with the registered hash. If the values match, the document's contents are consistent with the registered version. A mismatch indicates that the document may have changed. This mechanism verifies document integrity but does not independently prove that the original certificate is genuine or that the material satisfies the required quality standards. Those aspects depend on trusted manufacturers, authorized inspectors, and appropriate testing procedures.

```
function registerQualityCertificate(
    uint256 materialId,
    bytes32 certificateHash
) external {
    require(
        materialExists[materialId],
        "Material does not exist"
    );

    require(
        registeredManufacturers[msg.sender] ||
        msg.sender == materials[materialId].currentCustodian,
        "Not authorized"
    );

    require(
        certificateHash != bytes32(0),
        "Invalid certificate hash"
    );

    materials[materialId].certificateHash = certificateHash;
}
```

5.2.5 Recording Material Inspection Results

Quality inspection is an important part of construction material verification. Authorized inspectors examine materials using the applicable testing procedures and record the results against the corresponding Material ID. The inspection function checks the inspector's authorization and records the submitted inspection outcome. Depending on the implemented workflow, the status may be recorded as Pending, Passed, or Failed.

For example, an inspector can record the inspection result for STEEL-2026-001. Authorized stakeholders can then retrieve the recorded status and review the associated inspection history. Inspection updates should be recorded as new blockchain transactions so that the sequence of events remains available for auditing. Blockchain preserves the submitted records, but the accuracy of the results depends on the reliability of the inspection process.

```
mapping(address => bool) public authorizedInspectors;

function authorizeInspector(address inspector) external {
    require(
        registeredManufacturers[msg.sender],
        "Only registered manufacturers can authorize"
    );

    require(
        inspector != address(0),
        "Invalid inspector address"
    );

    authorizedInspectors[inspector] = true;
}

function recordInspectionResult(
    uint256 materialId,
    string memory inspectionResult
) external {
    require(
        materialExists[materialId],
        "Material does not exist"
    );

    require(
        authorizedInspectors[msg.sender],
        "Not an authorized inspector"
    );

    bytes32 resultHash = keccak256(bytes(inspectionResult));

    require(
        resultHash == keccak256(bytes("Pending")) ||
        resultHash == keccak256(bytes("Passed")) ||
        resultHash == keccak256(bytes("Failed")),
        "Invalid inspection result"
    );

    materials[materialId].inspectionStatus = inspectionResult;

    if (resultHash == keccak256(bytes("Failed"))) {
        materials[materialId].materialStatus = "Rejected";
    }
}
```

5.2.6 Transferring Materials Through the Supply Chain

BuildChain provides a mechanism for recording material transfers between manufacturers, suppliers, distributors, and contractors.The transferMaterial() function can be designed to validate the sender's authorization, confirm that the material exists, check the recipient's address, and verify that the batch is eligible for transfer.

When the transaction satisfies the required conditions, the contract updates the recorded custodian and stores the transfer event. The previous transfer records remain part of the material's transaction history.For example, the movement of STEEL-2026-001 may follow this sequence:

Manufacturer → Supplier → Distributor → Contractor

Each authorized transfer contributes to the chronological blockchain history. This enables stakeholders to review the recorded movement of a material batch and identify inconsistencies in its transfer records.

```
function transferMaterial(
    uint256 materialId,
    address newCustodian
) external {
    require(
        materialExists[materialId],
        "Material does not exist"
    );

    require(
        materials[materialId].currentCustodian == msg.sender,
        "Not the current custodian"
    );

    require(
        newCustodian != address(0),
        "Invalid recipient address"
    );

    require(
        keccak256(
            bytes(materials[materialId].materialStatus)
        ) != keccak256(bytes("Rejected")),
        "Rejected material cannot be transferred"
    );

    require(
        keccak256(
            bytes(materials[materialId].materialStatus)
        ) != keccak256(bytes("Delivered")),
        "Delivered material cannot be transferred"
    );
```

```
        materials[materialId].currentCustodian = newCustodian;

        materialHistory[materialId].push(newCustodian);
    }
```

5.2.7 Verifying Material Authenticity

The verifyMaterial() function retrieves the registered material information and checks whether the supplied Material ID corresponds to an existing record.Depending on the implemented verification rules, the system can check:

- Whether the Material ID is registered.
- Whether the manufacturer details are available.
- Whether the certificate reference or hash matches the registered record.
- Whether the required inspection result has been recorded.
- Whether the material status permits acceptance.
- Whether the recorded transfer history is available.

Each material can be associated with a unique QR code containing its Material ID or a verification link. When a contractor scans the QR code, the web application retrieves the corresponding blockchain information and displays the registered details and verification status.

Possible outcomes include Verified, Pending Inspection, Inspection Failed, Certificate Mismatch, and Record Not Found, provided the relevant checks are implemented.

A QR code alone cannot prove physical authenticity because it can be copied onto counterfeit materials. Therefore, reliable verification also requires trusted registration, appropriate physical identification, and authorized inspection.

```
    function verifyMaterial(uint256 materialId)
        external
        view
        returns (bool)
    {
        require(
            materialExists[materialId],
            "Material does not exist"
        );

        Material storage material = materials[materialId];

        require(
            material.manufacturer != address(0),
            "Manufacturer not recorded"
        );
```

```
        require(
            material.certificateHash != bytes32(0),
            "Certificate not registered"
        );

        require(
            keccak256(bytes(material.inspectionStatus))
                == keccak256(bytes("Passed")),
            "Inspection not passed"
        );

        require(
            keccak256(bytes(material.materialStatus))
                != keccak256(bytes("Rejected")),
            "Material rejected"
        );

        return true;
    }
```

5.2.8 Deploying and Integrating with a Frontend

The getMaterialHistory() function is designed to provide access to the recorded events associated with a material batch.The history may include registration, certificate references, inspection results, transfers, and delivery confirmation. These events can be associated with blockchain transaction records and timestamps.

Authorized contractors and inspectors can review this history before accepting materials at a construction site. The historical record also supports audits and investigations when a material-related problem is identified.The material record should remain available after verification or transfer. Instead of deleting the record, the system preserves its history and updates its status through authorized transactions.

5.3 Workflow:

The workflow of BuildChain, illustrated in Figure 5.1, describes the interactions among manufacturers, quality inspectors, suppliers, distributors, contractors, and the blockchain network.

The process begins when an authorized manufacturer registers a construction material batch and assigns a unique Material ID. The manufacturer provides the relevant material details and associates the available certificate reference with the record. An authorized inspector then records the inspection outcome after the required checks have been performed.

When the material is eligible for transfer, the current authorized custodian initiates a transfer to the next participant. The smart contract validates the transaction

according to its predefined rules and records the transfer event. This process continues as the material moves through the supply chain.When the material reaches the construction site, the contractor scans its QR code. The application retrieves the corresponding registered material information and displays the available certificate details, inspection status, and transfer history.

If the material has a valid registered record and satisfies the implemented verification rules, the application displays the corresponding status. If the Material ID is not registered, a certificate hash does not match, or a required inspection has failed, the system displays an appropriate warning for further investigation.Finally, the authorized receiving participant records delivery or receipt confirmation. This event is added to the blockchain history, allowing authorized stakeholders to review the recorded sequence of material-related transactions.

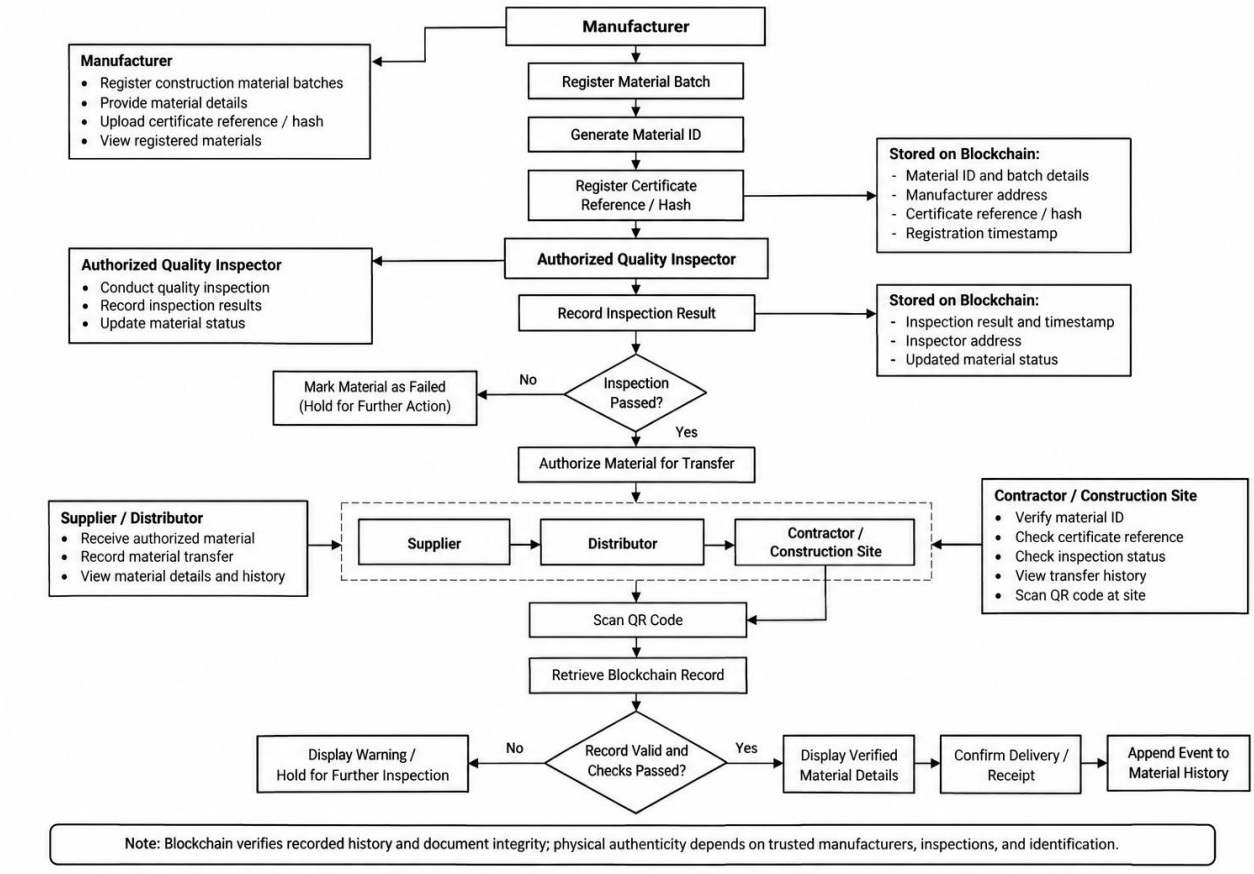


Figure 5.1: Workflow of BuildChain—Blockchain-Based Construction Material Authentication and Traceability System

The workflow illustrates how BuildChain combines blockchain transactions, smart contracts, QR-code identification, and authorized inspections to support construction material authentication and traceability.

The system's actual capabilities depend on the smart contract functions and frontend features that are implemented and tested. Features such as certificate-hash comparison, role-based authorization, QR-code verification, and complete history retrieval should be reported as implemented only after successful testing.

6 BENEFITS

Blockchain technology enhances construction material authentication and traceability by introducing transparency, security, and accountability across the supply chain. Traditional construction material management often relies on paper-based certificates, separate databases, and manual verification processes, which can lead to counterfeit materials, document manipulation, inefficient tracking, and difficulties in identifying the origin of materials. By leveraging blockchain technology and smart contracts, BuildChain provides a shared, tamper-resistant record of material registration, quality inspection, and supply-chain transfers. The following are the key benefits of the proposed system.

6.1 Enhanced Security and Immutability

Blockchain maintains a chronological record of material registration, certificate references, inspection results, and transfer transactions. Once transactions are confirmed on the blockchain, altering previously recorded information becomes difficult without detection. This helps reduce the risk of unauthorized modification of material records and supports reliable auditing. However, blockchain preserves submitted information; the correctness of the original data depends on authorized participants and reliable inspection procedures.

6.2 Improved Construction Material Authentication

BuildChain assigns a unique Material ID to each registered construction material batch. This identifier can be associated with a QR code that enables stakeholders to retrieve the corresponding blockchain record. By checking the registered manufacturer, certificate hash, inspection status, and material status, the system helps identify inconsistencies in recorded information. This improves the verification process and can help detect suspicious or unregistered material batches. Physical authenticity still requires suitable identification methods and trusted verification procedures because QR codes can be copied.

6.3 Transparency and Auditability

Blockchain provides a shared record of material-related transactions for participating manufacturers, quality inspectors, suppliers, distributors, and contractors. Authorized stakeholders can review the recorded material details, inspection outcomes, and custody history. This reduces dependence on separately maintained records and helps stakeholders investigate discrepancies, clarify responsibilities, and review the sequence of material movements.

6.4 Improved Supply-Chain Traceability

Construction materials may pass through several participants before reaching a construction site. BuildChain records changes in the current custodian and maintains a history of recorded custody transfers. This allows stakeholders to trace a material batch from

its registered manufacturer through the supply chain to the receiving contractor. Better traceability can help identify missing records, unexpected transfers, and inconsistencies in the documented movement of materials.

6.5 Efficient Certificate Management

Quality certificates, laboratory reports, and inspection documents are important for verifying construction materials. BuildChain can associate a cryptographic hash of a certificate with the corresponding material record. When the document is retrieved, its calculated hash can be compared with the registered hash to determine whether its contents match the recorded version. The document itself may be stored separately, while the blockchain maintains the reference hash. This supports document-integrity verification but does not independently establish that a certificate is genuine or that the material meets the required standards.

6.6 Smart Contracts for Automated Validation

Smart contracts can enforce predefined rules for material registration, inspector authorization, inspection-result recording, and material transfers. For example, a contract can reject a transfer request when the material record does not exist, the caller is not the current custodian, or the material has been marked as rejected. Automating these checks reduces reliance on manual validation and promotes consistent application of the system's rules. The effectiveness of this benefit depends on the rules actually implemented in the smart contract.

6.7 Improved Quality Inspection and Accountability

BuildChain associates inspection outcomes with the relevant Material ID. Authorized inspectors can record whether a material batch is pending inspection, has passed inspection, or has failed inspection. These records help contractors and other stakeholders review the recorded inspection status before accepting a material batch. Maintaining inspection information alongside material records improves accountability and supports the investigation of quality-related issues. The reliability of the recorded result still depends on the inspector's competence, the testing procedure, and the accuracy of the submitted information.

6.8 Reduced Fraud and Unauthorized Record Changes

Construction material fraud may involve altered certificates, false material identities, or inconsistent supply-chain records. BuildChain makes recorded information easier to verify and makes unauthorized changes to confirmed blockchain records more difficult. Smart-contract access controls can also restrict selected operations to authorized manufacturers, inspectors, or current custodians. These mechanisms can help identify suspicious activity and discourage certain forms of record manipulation. They cannot

eliminate fraud entirely, especially when false information is submitted initially or physical materials are substituted after registration.

6.9 Faster Verification and Better Decision-Making

Using a QR code linked to a Material ID, a contractor can retrieve the corresponding blockchain record through the web application. The application can display available material details, certificate information, inspection status, and recorded transfer history in one place. This can reduce the time spent searching through paper documents or contacting multiple supply-chain participants. Faster access to recorded information supports more informed material acceptance and helps stakeholders identify records that require further investigation.

6.10 Improved Documentation and Reduced Paperwork

BuildChain maintains material-related information digitally, reducing dependence on paper-based records and repeated manual documentation. Material registration, certificate references, inspection outcomes, and transfer records can be accessed through the application, subject to the system's access design. This can simplify record retrieval, improve documentation consistency, and reduce administrative effort. Digital storage and blockchain transactions still require appropriate infrastructure, maintenance, and data-management practices.

Implementing BuildChain can improve the transparency, traceability, and verifiability of construction material records throughout the supply chain. By combining blockchain, smart contracts, certificate-hash verification, and QR-code-based record retrieval, the proposed system can help stakeholders identify inconsistencies, review inspection results, and track the recorded movement of material batches. Its effectiveness depends on reliable data entry, authorized inspections, appropriate access controls, and suitable physical identification measures. Blockchain should therefore be considered a supporting mechanism for material authentication and traceability rather than an independent guarantee of physical material quality or authenticity.

7 CHALLENGES

While blockchain-based construction material authentication and traceability systems offer several advantages, they also present technical, operational, and security challenges that must be considered for effective implementation. The reliability of BuildChain depends not only on blockchain technology but also on accurate data entry, trusted quality inspections, secure participant access, and proper integration with the construction supply chain. The following are some of the key challenges.

7.1 Scalability and Transaction Costs

Construction supply chains may involve large numbers of material batches, inspection updates, certificate registrations, and transfer transactions. Recording every event on a blockchain can increase storage requirements, transaction processing time, and deployment or execution costs. Public blockchain networks may also experience congestion and fluctuating gas fees. To address these issues, BuildChain could store essential verification data or certificate hashes on-chain while keeping large documents and supporting records in suitable off-chain storage.

7.2 Reliability of Submitted Information

Blockchain preserves confirmed records, but it cannot independently determine whether the information entered by a manufacturer, supplier, or inspector is true. Incorrect batch details, false certificates, or inaccurate inspection results may be recorded permanently if they are accepted by the system. Therefore, BuildChain requires reliable participant identification, authorized inspections, appropriate testing procedures, and validation of submitted information. The system should be treated as a way to improve record integrity rather than as an independent guarantee of material authenticity.

7.3 Physical Counterfeiting and QR-Code Duplication

BuildChain can associate a QR code with a registered Material ID, allowing users to retrieve the corresponding blockchain record. However, counterfeiters may copy a genuine QR code and attach it to a different or inferior material batch. A valid blockchain record therefore does not necessarily prove that the physical material being inspected is the material originally registered. Additional measures, such as tamper-evident labels, secure batch markings, trusted inspection procedures, or suitable physical identification technologies, may be required to strengthen verification.

7.4 Privacy and Access Control

Construction supply-chain records may contain commercially sensitive information, including supplier identities, delivery details, inspection reports, and transaction histories. Making all information publicly accessible may create privacy and

business-confidentiality concerns. BuildChain must define which information can be viewed by different participants and which operations require authorization. Sensitive documents can be stored off-chain with suitable access controls, while only necessary hashes and transaction references are recorded on the blockchain.

7.5 Smart Contract Security and Maintenance

Smart contracts control important operations such as material registration, inspector authorization, inspection-result recording, and material transfers. Programming errors or incorrectly defined access permissions may allow unauthorized operations or prevent legitimate transactions. Once a contract is deployed, modifying its logic may also require redeployment or a carefully designed upgrade mechanism. Thorough testing, access-control checks, code review, and security auditing are therefore important before the system is used in a real construction supply chain.

7.6 Integration with Existing Supply-Chain Processes

Manufacturers, suppliers, distributors, laboratories, and contractors may already use different databases, inventory systems, and documentation procedures. Integrating BuildChain with these existing systems can require additional software development, data conversion, staff training, and coordination between organizations. Differences in data formats and operating procedures may also cause inconsistencies. Successful implementation requires common data standards and cooperation among participating stakeholders.

7.7 Dependence on Internet Connectivity and Infrastructure

Blockchain-based verification and transaction submission generally require access to the blockchain network, while QR-code scanning and record retrieval may depend on a connected web application. Construction sites in remote locations or areas with unreliable internet connectivity may experience delays when checking records or submitting updates. Suitable network infrastructure, reliable application hosting, and carefully designed offline data-capture procedures may help reduce these difficulties. Any offline records should be validated before being accepted as confirmed blockchain transactions.

7.8 Legal and Regulatory Considerations

Construction materials must comply with applicable building regulations, quality standards, certification requirements, and inspection procedures. Blockchain records do not automatically establish legal compliance or replace certificates issued by recognized authorities. Organizations implementing BuildChain must ensure that their verification process follows relevant regulations and that the recorded information can be used appropriately for audits and dispute resolution. Requirements may vary according to the material type, location, and applicable standards.

8 CONCLUSION

This report explores the integration of blockchain technology into construction material authentication and traceability, addressing key challenges in traditional construction supply-chain management. Conventional processes often rely on paper-based certificates, separate databases, and manual verification, which can create opportunities for document manipulation, counterfeit material substitution, delays in verification, and difficulties in tracking materials across multiple participants. By leveraging blockchain's immutability, shared recordkeeping, and transparency, the proposed BuildChain system provides a structured approach to maintaining reliable records of material registration, quality inspection, and supply-chain transfers.

The proposed system uses smart contracts to manage material registration, associate certificate hashes with material records, record inspection results, and update the current custodian during transfers. A unique Material ID linked to a QR code allows stakeholders, including suppliers, distributors, and contractors, to retrieve the corresponding material information and review its recorded history. These features can improve traceability, simplify access to material records, and help stakeholders identify inconsistencies before accepting construction materials.

BuildChain demonstrates how blockchain can support more accountable construction material management by maintaining a chronological record of submitted transactions. However, blockchain does not independently prove the physical authenticity or quality of construction materials. The reliability of the system depends on accurate data entry, authorized inspections, trustworthy certificate sources, and appropriate physical identification measures. Therefore, the proposed approach should complement, rather than replace, established material testing and quality-assurance procedures.

8.1 Future Outlook for Enhancements

To further improve BuildChain, several enhancements can be explored. Integrating Internet of Things (IoT) technologies and suitable identification devices could strengthen the association between physical material batches and their digital records. For example, tamper-evident labels, secure identifiers, or compatible scanning devices could help reduce the risk of QR-code duplication and material substitution. Such mechanisms would require appropriate validation and would not eliminate the need for physical inspection.

Artificial intelligence (AI) and data analytics could also be incorporated to identify unusual supply-chain patterns, inconsistent inspection records, repeated certificate references, or suspicious transfer activities. These tools could help prioritize records for investigation and support quality-control decisions. Their effectiveness would depend on the availability of reliable data and suitable validation.

Another possible enhancement is the development of a more comprehensive digital certificate-management module. This module could calculate certificate hashes, compare them with registered blockchain hashes, maintain document references, and provide clear verification outcomes. Secure off-chain storage could be used for large documents, while the blockchain would retain the corresponding hashes and essential transaction records. Access controls would be needed to protect confidential information.

Scalability and interoperability could be improved by integrating BuildChain with existing inventory-management systems, laboratory information systems, and construction project-management applications. For larger deployments, permissioned blockchain networks or suitable scaling solutions could be evaluated to improve transaction throughput, manage access permissions, and control operational costs. These choices would need to be assessed according to the number of participants, transaction volume, privacy requirements, and deployment environment.

Finally, future versions could provide timestamped inspection and transfer histories, receiving-party confirmation, role-based access control, and comprehensive audit reports. A user-friendly web application could allow authorized stakeholders to scan QR codes and review material details, certificate-integrity results, inspection status, and custody history through a single interface.

By incorporating these enhancements and validating them through systematic testing, BuildChain could evolve into a more comprehensive construction material traceability solution. The proposed system illustrates the potential of blockchain and smart contracts to improve record integrity, transparency, and accountability throughout the construction supply chain, while reliable inspections and physical verification remain essential for ensuring material authenticity and quality.

9 SDG's ADDRESSED

The BuildChain system aligns with two United Nations Sustainable Development Goals (SDGs) by promoting technological innovation in construction infrastructure and responsible construction material management. The proposed system uses blockchain technology, smart contracts, certificate-hash verification, and QR-code-based material identification to improve transparency and traceability throughout the construction supply chain.

SDG 9: Industry, Innovation, and Infrastructure

BuildChain supports SDG 9 by applying blockchain technology to improve the traceability and reliability of construction material records. The system records material registration, manufacturer details, quality inspection results, certificate references, and supply-chain transfers. Smart contracts can enforce predefined rules for material registration and transfers, while QR-code-based verification enables stakeholders to retrieve the corresponding material records. These features promote technological innovation and support more transparent and accountable construction infrastructure management.

SDG 12: Responsible Consumption and Production

BuildChain supports SDG 12 by improving construction material traceability and supporting informed material acceptance decisions. Stakeholders can review registered material details, certificate hashes, inspection results, and recorded transfer history to identify inconsistencies. This can help reduce the risk of unsuitable or counterfeit materials entering construction projects and potentially prevent avoidable rework and material waste. Digital recordkeeping can also reduce dependence on paper-based documentation. These benefits depend on accurate records, reliable inspections, and appropriate physical material identification.

By addressing SDG 9 and SDG 12, BuildChain demonstrates how blockchain-based traceability can support innovation in construction infrastructure and responsible construction material management. Blockchain helps preserve recorded information, while trusted certification, physical inspection, and appropriate testing remain essential for establishing material authenticity and quality.

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11 APPENDIX A

https://github.com/katakamsirivalli/Blockchain_Project

https://youtu.be/l5mWAtqd-hQ

https://drive.google.com/drive/folders/18gbVD_jK0A34hihZNLJfTmmEpT9wpBg6?usp=sharing

